

Coach Kai Deployment Status Report

Current Situation



What's Working

- **Enhanced Coach Kai Intelligence:** GPT-4o API integration is fully implemented
- **Function Calling:** Calendar, messaging, drill suggestions, and analysis tools are configured
- **PTT (Push-to-Talk):** Text-only Coach Kai with voice input is coded and ready
- **API Routes:** All `/api/coach-kai/*` endpoints are functional
- **Knowledge Base:** Pickleball deficiency-to-drill mapping is complete



What's Not Working

- **Production Deployment:** Stuck on old version with HeyGen avatar trying to load
 - **GitHub Push Blocked:** Exposed Twilio SID preventing code updates
 - **Coach Kai UI:** Shows “Starting up...” with “Video unavailable” error
 - **Latest Code Not Deployed:** Text-only Coach Kai with PTT not live on mindfulchampion.com
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Why Coach Kai is Stuck

The Problem

Your production deployment (mindfulchampion.com) is showing an **older version** of Coach Kai that used **HeyGen video avatars**. The screenshots show:

-  K (avatar placeholder)
-  Starting up...
-  Video unavailable. Text chat is always free!

What Happened

1. **December 31, 2025:** HeyGen avatar was intentionally removed (commit `dbe3303`)
2. **January 1, 2026:** Replaced with text-only Coach Kai with PTT feature
3. **Push Blocked:** Twilio SID exposure prevents pushing new code to GitHub
4. **Vercel Can't Update:** Without new commits on GitHub, Vercel can't deploy the latest code

The Current Code State

Local Repository (✅ Updated):

- components/coach/text-coach-kai.tsx (with PTT)
- app/api/coach-kai/chat/route.ts (GPT-4o intelligence)
- lib/coach-kai/enhanced-system-prompt.ts (emotional intelligence)

GitHub Repository (🔴 Outdated):

- Still has commit with exposed Twilio SID
- Can't accept new pushes due to security block
- Last successful push: December 31

Vercel Deployment (🔴 Outdated):

- Showing old HeyGen avatar interface
- Stuck on "Starting up..." because HeyGen component was removed
- Can't load text-coach-kai.tsx because it's not deployed

Fix Plan (20 Minutes Total)

Phase 1: Unblock GitHub (5 minutes)

Option A: Rotate Twilio SID (Recommended)

1. Go to <https://console.twilio.com>
2. Create new subaccount → Get new SID
3. Update Vercel environment variables with new SID
4. Visit GitHub unblock link and click "I have rotated this secret"
5. Push code: `cd /home/ubuntu/mindful_champion && git push origin master`

Option B: Remove from Git History (Advanced)

```
cd /home/ubuntu/mindful_champion
echo "AC17eddale6d788cf548dc5d99300b2a66" > secrets.txt
wget https://repo1.maven.org/maven2/com/madgag/bfg/1.14.0/bfg-1.14.0.jar
java -jar bfg-1.14.0.jar --replace-text secrets.txt .git
git reflog expire --expire=now --all
git gc --prune=now --aggressive
git push origin --force --all
```

Phase 2: Deploy to Vercel (5 minutes)

After GitHub is unblocked:

1. Push Latest Code:

```
bash
cd /home/ubuntu/mindful_champion
git status # Verify clean state
git push origin master
```

2. Verify Vercel Auto-Deploy:

- Go to <https://vercel.com/dashboard>
- Check for new deployment triggered by push
- Monitor build logs for any errors

3. If Manual Deploy Needed:

```
bash
cd /home/ubuntu/mindful_champion/nextjs_space
npx vercel --prod
```

Phase 3: Verify Coach Kai (5 minutes)

1. Visit Production Site:

- Open <https://mindfulchampion.com/train/coach>
- Should see text-only interface (no video avatar)
- Should see PTT microphone button

2. Test Features:

- Send a text message to Coach Kai
- Verify GPT-4o response
- Test PTT button (hold to speak)
- Check action cards (calendar, drills, etc.)

3. Check API Responses:

```
bash
curl -X POST https://mindfulchampion.com/api/coach-kai/chat \
-H "Content-Type: application/json" \
-d '{"messages": [{"role": "user", "content": "Hello"}]}'
```

Phase 4: Monitor Production (5 minutes)

1. Check Vercel Logs:

- Functions tab for errors
- Real-time logs for API calls
- Performance metrics

2. Test on Mobile:

- iOS Safari (your screenshot device)
- Android Chrome
- iPad

3. Verify Environment Variables:

```
✓ DATABASE_URL
✓ NEXTAUTH_SECRET
✓ ABACUSAI_API_KEY (for GPT-4o)
✓ ELEVENLABS_API_KEY (optional for TTS)
✓ TWILIO_ACCOUNT_SID (new one after rotation)
✓ TWILIO_AUTH_TOKEN
```

Expected Outcome

Before Fix

- Production: HeyGen avatar stuck on "Starting up..."
- GitHub: Push blocked by security alert
- Coach Kai: Non-functional, shows error

After Fix

- ✓ Production: Text-only Coach Kai with PTT button
- ✓ GitHub: Unblocked, accepts new pushes
- ✓ Coach Kai: Fully functional with GPT-4o intelligence
- ✓ Features: Action cards, drill suggestions, calendar integration

Technical Details: What's in the New Coach Kai

Enhanced Intelligence (lib/coach-kai/enhanced-system-prompt.ts)

- **Model:** GPT-4.1-mini (Abacus.AI endpoint)
- **Emotional Intelligence:** Detects frustration, motivation, goals
- **Context Awareness:** User skill level, recent matches, active goals
- **Personality:** Encouraging mentor with pickleball expertise

Function Calling (lib/coach-kai/function-handler.ts)

```
FUNCTION_TOOLS = [
  {
    type: "function",
    function: {
      name: "add_to_calendar",
      description: "Add tournaments, practices, or events to user's calendar",
      parameters: { event_name, date, time, location, notes }
    }
  },
  {
    type: "function",
    function: {
      name: "send_message",
      description: "Connect user with partners or coaches",
      parameters: { recipient_id, message }
    }
  },
  {
    type: "function",
    function: {
      name: "suggest_resource",
      description: "Recommend drills, goals, or training programs",
      parameters: { type, title, details, why_it_helps }
    }
  },
  {
    type: "function",
    function: {
      name: "analyze_technique",
      description: "Provide video analysis insights",
      parameters: { video_url, focus_areas, feedback }
    }
  }
]
```

PTT (Push-to-Talk) Feature

- **Web Speech API:** Browser-native voice recognition

- **Real-time Transcription:** Shows what Coach Kai hears
- **Mobile Optimized:** Works on iOS Safari and Android Chrome
- **Visual Feedback:** Animated microphone icon during recording

Knowledge Base (Pickleball Deficiencies)

- **Grip Issues** → Drill: “Grip Strengthening Exercises”
 - **Footwork Problems** → Drill: “Split-Step Timing Practice”
 - **Serve Consistency** → Drill: “Target Serve Training”
 - **Dink Control** → Drill: “Soft Hands Dinking”
 - **Third Shot Drop** → Drill: “Drop Shot Consistency”
 - **Overhead Smash** → Drill: “Power Smash Practice”
 - **Volleys** → Drill: “Reflex Volley Drills”
 - **Court Positioning** → Drill: “Kitchen Line Positioning”
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Troubleshooting

If GitHub Push Still Blocked

1. Verify you clicked “I have rotated this secret” in the GitHub alert
2. Try pushing to a new branch first:

```
bash
git checkout -b deployment-fix
git push origin deployment-fix
```

3. If still blocked, contact GitHub support with the alert URL

If Vercel Build Fails

1. **Check Environment Variables:** All required vars must be set
2. **Check Build Logs:** Look for specific error messages
3. **Common Issues:**
 - Missing `ABACUSAI_API_KEY` → Coach Kai won’t work
 - Missing `DATABASE_URL` → Authentication fails
 - Missing `NEXTAUTH_SECRET` → Sessions break

If Coach Kai Still Shows “Starting up...”

1. **Hard Refresh:** Cmd+Shift+R (Mac) or Ctrl+Shift+R (Windows)
2. **Clear Cache:** Browser settings → Clear site data
3. **Check Deployment:** Verify latest commit hash on Vercel matches GitHub
4. **Inspect Console:** F12 → Console tab for JavaScript errors

If PTT Button Not Working

1. **Check Browser Permissions:** Allow microphone access
 2. **HTTPS Required:** PTT only works on secure connections
 3. **Supported Browsers:** Chrome, Safari, Edge (not Firefox yet)
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Files Changed in Latest Code

New Files

```
lib/coach-kai/
├─ enhanced-system-prompt.ts (GPT-4o prompt with emotional intelligence)
├─ function-handler.ts (action detection and processing)
├─ knowledge-base.ts (deficiency-to-drill mapping)

components/coach/
├─ text-coach-kai.tsx (PTT-enabled text interface)
```

Removed Files

```
components/coach/
├─ heygen-coach-kai.tsx (deleted, backed up as .bak)

app/api/heygen/
├─ token/route.ts (deleted)
├─ cleanup/route.ts (deleted)
```

Updated Files

```
app/train/coach/page.tsx (imports TextCoachKai instead of HeyGenCoachKai)
app/api/coach-kai/chat/route.ts (uses enhanced prompt and function calling)
app/api/coach-kai/execute-action/route.ts (processes confirmed actions)
```

Deployment Checklist

Pre-Deployment

- ☐ Twilio SID rotated or git history cleaned
- ☐ GitHub push unblocked
- ☐ Vercel environment variables verified
- ☐ Local build tested: `npm run build`

Deployment

- ☐ Code pushed to GitHub
- ☐ Vercel deployment triggered
- ☐ Build completed successfully
- ☐ No errors in Vercel function logs

Post-Deployment

- ☐ Coach Kai loads without “Starting up...” error
- ☐ Text input works and gets GPT-4o responses
- ☐ PTT button visible and functional
- ☐ Action cards appear for drills/calendar/messages
- ☐ Mobile devices (iOS/Android) working
- ☐ API response times acceptable (<5 seconds)

Success Metrics

Performance Targets

- **First Response:** <3 seconds
- **Streaming Speed:** 20-50 tokens/second
- **Action Cards:** Appear within 1 second of response
- **PTT Transcription:** Real-time (<500ms delay)

User Experience

- **No Video Placeholder:** Text-only interface loads instantly
 - **Clear Instructions:** Help text visible for first-time users
 - **Smooth Animations:** Framer Motion transitions polished
 - **Responsive Design:** Works on all screen sizes
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Next Steps After Fix

1. **Monitor for 24 Hours:** Watch Vercel logs for errors
 2. **Collect User Feedback:** Test with real users
 3. **Optimize Prompts:** Fine-tune system prompt based on responses
 4. **Add Features:**
 - Voice output (ElevenLabs TTS)
 - Image upload for technique analysis
 - Multi-language support
 - Session history/memory
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Support Resources

- **Vercel Dashboard:** <https://vercel.com/dashboard>
- **GitHub Repository:** <https://github.com/SilentVector001/mindful-champion>
- **Twilio Console:** <https://console.twilio.com>
- **Abacus.AI:** <https://apps.abacus.ai>

Need Help? The code is ready to deploy. Just need to:

1. ☒ Rotate Twilio SID
2. ☒ Unblock GitHub
3. ☒ Push and deploy

Estimated Time to Live: 20 minutes ⚡